

Chapter 8

Bitcoin Manual for Dummies

For the final chapter of this book, the author would like to present a broad and elementary overview of the practical application of Bitcoin. Understanding the Bitcoin whitepaper's breakthrough, its monetary properties, the biblical context, and seeing its real-world efficacy paves the foundation for understanding its role and effectiveness against the economic significance of the mark of the beast. Still, knowledge alone is not sufficient to constitute a solution without application.

One will never receive the benefit of a tool without learning to use it, as looking under the hood of a car and studying its engine components will never be as effective as manoeuvring the wheel and the pedals to get from one location to another. Whilst the author will not go into the greatest detail on every component under the hood, this final chapter will, in a most understandable way, explain the core components of Bitcoin and their respective applications for day-to-day use of storing and transacting value. Kind of like a user manual of Bitcoin for dummies.

That said, the information within this chapter is non-authoritative, only for reference, and provides a birds-eye overview to the nuances of practically using Bitcoin. Every part of Bitcoin's application (and in fact, any technology for this matter) is subjective and circumstantial and should not be taken as gospel. And like any other manual, it can be read through from start to finish to provide a bird's-eye overview, but may be most useful when consulted selectively, as specific needs arise.

Storing & Self-Custodying Bitcoin

The word "Bitcoin" officially refers to two things:

- 1) The Bitcoin software protocol: Bitcoin
- 2) The monetary unit within Bitcoin's software protocol:
 - Singular: 1 bitcoin, 1 BTC
 - Plural: 2 bitcoins, 2 BTC

Being an owner and self-custodying your bitcoins means true freedom from banks' control, government restrictions, and the degradation of fiat currencies. Bitcoin recovers back all the power you should have over your money, but that also means no one else is responsible for your money, except yourself. With full sovereignty also comes full responsibilities.

To be able to possess bitcoins and use them for future transactions, you must first have a Bitcoin wallet. A Bitcoin wallet is how you interact with the Bitcoin network. Consider how you would select a Bitcoin wallet; it is analogous to how you would like to store your physical gold and silver. A small purse should suffice for a small enough amount to carry for interacting with

merchants and purchasing daily necessities, while a larger vault should remain disconnected from the outside world for storing larger amounts of one's funds.

This section will go through what kind of Bitcoin wallets there are, the utility of different types of wallets, good practices, and common pitfalls of using your own wallet to self-custody bitcoins. Note that Bitcoin is NOT stored in your wallet, your private key is stored in your wallet, giving you access to your bitcoins on the Bitcoin blockchain.

Key words

- **Custodial:** *A third party (like a wallet provider) holds the keys for you as an intermediary. You do not self-custody your own funds when you deposit bitcoins into a custodial wallet.*
- **Non-custodial:** *You alone hold the private keys. No bank intermediary is between you and your funds, but if you lose your keys, there is no customer support.*
- **Private key:** *A secret, 256-bit number that allows you to spend bitcoins from the wallet. If someone has your private key, they have your bitcoins. In a wallet, this is usually hidden behind a seed phrase.*
- **Seed phrase (recovery phrase):** *A list of 12 or 24 random words that acts as a human-readable private key. It can recreate your entire wallet and all its addresses if your wallet app/device is lost, hence also known as a recovery phrase. Once you commit your seed phrase to memory, your brain essentially becomes also a "wallet".*
- **Public key:** *A large number mathematically derived from the private key. It is used to generate addresses for receiving, storing, and sending your bitcoins.*
- **Wallet address(es):** *A shorter, hashed version of your public key (e.g., bc1q...). Think of this as your "mailbox" address. You give this to others so they can send you bitcoins.*
- **Wallet password/ pin:** *The password to access the wallet, similar to your phone's password to prevent strangers from immediately accessing your device in case of loss or theft. You do not lose your funds if you forget the pin.*
- **Passphrase:** *An optional phrase (like "13th" or "25th" word to the seed phrase) that adds a new wallet as an additional layer of protection to your funds, but if you forget the passphrase, you lose all the funds in that passphrase wallet.*
- **UTXOs (Unspent Transaction Outputs):** *Your wallet is a collection of "digital chunks" of bitcoins (UTXOs) from previous transactions. Think of it as nuggets of gold varying in weights in your pouch.*

Software wallets

Software wallets, known also as hot wallets or soft wallets, are wallets that are connected to the internet and exclusive as a software on your device.

Typical attributes of software wallets being:

Pros	Cons
- Easy to setup - Convenient to use - User friendly - Free to download - Can recovery one's refunds with wallet provider if custodial	- Greater surface of attack, susceptible to hacking, phishing, malware, and other cyber threats - Requires trust if custodial

Software wallets are ideal for storing insignificant amounts of BTC for everyday transactions, akin to having \$200 of cash in your physical wallet. But it is not recommended to be used to store a significant amount of bitcoins for the user. Users typically install software wallets on their mobile phones as mobile, carry-on wallets used in tandem with Lightning payments (which we will soon cover).

It is the author's recommendation to prioritise non-custodial over custodial wallets if you desire to maximise the sovereignty of your funds, as you will retain your private key as a seed phrase. However, if you are not yet confident to fully custody your funds or find the need to protect a seed phrase for a relatively small quantity of funds inconvenient, custodial software wallets are also fine for starting off. In the Freedom Convoy's story from the previous chapter, when the Canadian court requested the non-custodial wallet provider Nunchuk to freeze the funds of the protesters, the following was the company's reply (1):

We do not collect any user identification information... We cannot freeze our users' assets. We cannot prevent them from being moved. We have no knowledge of the existence, nature, value and location of our users' local assets. This is by design.

When choosing a software wallet for use, the author highly recommend choosing wallets that satisfy the following:

- **Open-sourced:** The source code for the software is fully published on a public online platform like GitHub to ensure there are no "backdoors" in the software.
- **Bitcoin-only:** Bitcoin-only software minimises the attack surface of the software. Wallets that support thousands of random cryptocurrencies are less secure and more prone to bugs.
- **Lightning Integration:** Enables instantaneous payments of bitcoins from A to B, necessary for users who make payments from their software wallets.
- **Non-custodial:** You hold the seed phrase; the developers have zero knowledge, control, and access to your funds.

Software wallets can also exist on desktops, such as Sparrow Wallet, which is desktop-exclusive and considered the 'gold standard' by many Bitcoiners (this is not an endorsement). Some software wallets (especially desktop ones) can be used in tandem with a hardware wallet, in which case they serve as a user interface for the hardware wallet.

Hardware Wallets

Hardware wallets, known also as cold wallets or hard wallets, are wallets that are NOT connected to the internet and are almost exclusively non-custodial, programmed to randomly generate your recovery seed (containing your private key) without an internet connection.

Typical attributes of hardware wallets being:

Pros	Cons
- Significantly more secure - Users retain full control over keys and funds - Enables layered protection (passphrase)	- Costs money (between \$60 to \$200) - Less convenient for payments (trade off with greater security) - Not as beginner friendly as soft wallets - Requires careful backup of recovery seed

Hardware wallets are ideal for storing significant amounts of BTC for long-term savings. If a software wallet is your digital purse, then a hardware wallet is your digital vault. To use hardware wallets, users typically install a desktop software wallet to act as a software coordinator (user interface) for the hardware wallet.

When choosing a hardware wallet for purchase and use, the author highly recommends the following features and practices:

- **Open-sourced (or source-viewable):** The source code for the native software is fully published to ensure there are no “backdoors” in the devices’ software.
- **Bitcoin-only:** Bitcoin-only software minimises the attack surface of hardware’s official software.
- **Buy Direct:** Only buy directly from the manufacturer (e.g., Coinkite, Trezor, Blockstream). Never buy from Amazon or eBay, as the device could be tampered with.
- **Verify the Packaging:** Check for broken security seals. When you plug it in, most devices will run a “genuineness check” automatically via their official software.
- **Random Seed Generation:** If the device comes with a pre-loaded recovery seed that already has words printed on it, it is a scam. You must generate your own 12 or 24 words on the device itself.

Tutorials on wallets of your choice can be found online and on YouTube, a channel of recommendation is ‘BTCSessions’ by a passionate team of Bitcoin educators from Canada.

Private Key Management

If your Bitcoin wallet is a vault, the **private key** is the actual combination that unlocks it. However, because a raw private key looks like a terrifying string of 64 random letters and numbers, almost all modern wallets use a **seed phrase** (also known as a recovery phrase or

recovery seed). Your seed phrase grants you access to your funds when entered into any wallet, software or hardware. Keep it safe.

When you create a Bitcoin wallet, you will typically write down a 12 or 24 word seed backup. This seed phrase is a mnemonic form of your private key. Akin to the key to the front door of your home, whoever has the key has access to what it unlocks. Remember and protect your key at all costs.

While there have been a handful of Bitcoin thefts from third-party hacks over the decade, the overwhelming reported losses of bitcoins are due to people taking inadequate care, steps, and effort in remembering, protecting, and backing up their seed phrase. Self-custodying bitcoins grants you full sovereignty over your own funds, and that is the point as an uncensorable, unconfiscatable money, but you also bear all the responsibility for your funds. Unlike a bank account where the bankers hold your funds, there's no custodial recovery available for self-custodied wallets. Always remember:

Not your keys, not your coins.

Managing your private key is the most important responsibility in Bitcoin. Since there is no "Forgot Password" button in a non-custodial world, it is highly recommended to follow these "never-break" rules:

1. **Never “digitise” your seed:** Never take a photo or screenshot of your seed phrase, or save it in a Notes app, or email it to yourself, or store it in the cloud. Your seed phrase should **never** interact with the internet or any external device to the hardware wallet (that is the point of hardware wallets' design). Hackers use automated tools to scan for 12-word patterns in leaked databases. Keep it physical.
2. **The medium:** While your wallet might come with a cardboard “Recovery Card”, paper is fragile. It can be destroyed by fire, humidity, or even curious pets.
 - For soft wallets with insignificant amounts, paper backup stored in a secure, dry place is usually fine.
 - For hard wallets with more significant amounts, consider a stainless steel backup. These are small metal plates where you stamp. They are fireproof and waterproof, ensuring your keys survive a house fire. A simple DIY setup can be a nut, bolt, and 12 or 24 washers.
3. **Secure location:** Make sure to store your physical seed at a secure location that only the owner of the funds knows and has access to.
 - Make it simple but secretive. The harder it is to find your seed phrase, the easier also it is for you to forget where it is. There is a trade-off.
 - Don't split your 12 or 24 word seed phrase in separate locations. A good explanation to this is given by Andreas Antonopoulos (found in the Bitcoin Education Resources section).



4. **Commit to memory:** Arguably, long-term storage should be committed to long-term memory. One of the most powerful features of Bitcoin is that its seed phrase enables your brain to become a “wallet” once committed to memory, allowing anyone to take their bitcoins wherever they go (or flee to).
5. **Test recovery:** Before storing a significant amount of bitcoins in your wallet, it is good practice to store \$5 of bitcoin in your wallet first, and factory reset your wallet and “recover” it using your written seed phrase. If it works, you will know your backup is correct, so you don’t end up unknowingly storing an incorrect seed. Secondly, it demonstrates your funds are accessible on any wallet device using your seed phrase.
6. **Don’t store the seed phrase with passphrase at the same location:** See below.

Passphrases

The **passphrase** is one of the most misunderstood yet powerful tools within a Bitcoin wallet. You can think of it as the secret room in your house, or having a smaller vault within a vault. If your seed phrase is analogous to the key to your house, the passphrase is the secret door hidden behind the bookshelf. Optional, but allows even more security, and if you lose the “key” to your secret room, you get locked out of that room.

How It Works

When you add a passphrase to your seed phrase, the wallet’s software performs a mathematical calculation to create a completely new and different wallet.

The following are examples of how passphrases work with a seed phrase (do not use these common passphrases yourself). You will see how passphrases theoretically enable an infinite number of ‘secret rooms’ to your wallet. But, as a housekeeper, the more keys you have to manage, the more tedious it becomes.

Passphrase A	Apple Pie#123
Passphrase B	apple pie#123
Passphrase C	applo pio#123
Passphrase D	ApplePie#123

Seed phrase = Wallet

Seed phrase + Passphrase A = Wallet A

Seed phrase + Passphrase B = Wallet B

Seed phrase + Passphrase C = Wallet C

Seed phrase + Passphrase D = Wallet D

Contrary to the seed phrase, the hardware wallet doesn't "store" the passphrase. It simply calculates a new set of addresses based on whatever words you type in. If you type a different passphrase, you get a different wallet.

Why Use a Passphrase

There are two main reasons Bitcoiners use passphrases:

Protection against physical theft: If a thief finds your seed phrase written on paper in your drawer, they will immediately try to sweep the funds. If you have a passphrase, the thief will only see the original wallet. Your funds behind a passphrase wallet are invisible to them unless they also know that passphrase. Hence, don't store your seed phrase with your passphrase together!

Duress Protection: If you are ever forced to unlock your wallet (the \$5 wrench attack scenario), you can give the attacker the pin to your "main" wallet which holds a small amount of "decoy" BTC. They leave satisfied, never knowing a much larger passphrase wallet exists.

The Double-Edged Sword

WARNING! The passphrase is one of the most common way people lose their Bitcoin. Because it is so powerful, it is also dangerous. There is no "reset" for a passphrase. If you forget it, or even if you misspell a single letter or capitalise something differently, you will be looking at a completely different, empty wallet. Your funds will be mathematically impossible to recover.

If you decide to use a passphrase for your storage, take note of the following:

- **Case sensitivity:** "Apple" is not the same as "apple". Every character, space, and symbol must be exact.
- **Don't overcomplicate it:** You don't need a 50-character passphrase, a few memorable words together do the job well (e.g. nervous-llama-eats-potato).
- **Commit to memory:** Like the seed phrase, commit to memory, but also have a physical backup.

	Seed Phrase	Passphrase
Generated by	The hardware device (randomly)	You (your choice)
Store it on	Paper/metal backup (and your brain)	Separate backup (and your brain)
If Lost	Funds in the non-passphrase wallet are lost	Funds in the passphrase wallet are lost

Multi-Signature (Advanced)

So far, we have covered storing and self-custodying one's funds with a single private key, that is, a single signature (one signature per private key). Multi-signature (multi-sig) is akin to having a vault that requires two or more keys to open; it is a way of securing bitcoins where more than one signature is required to move funds. It uses a mathematical rule called M-of-N, the most common being 2-of-3, you have 3 keys total and you need any 2 of them to send a transaction.

The practice of multi-signature self-custody significantly increases the security of one's funds against external threats, but equally the complexity of self-custody. Therefore, a multi-sig setup may not be necessary for most individuals, as a single-seed-phrase wallet with passphrase protection already provides ample security. Here are a few benefits and reasons for adopting multi-sig. Remember, as Bitcoin grants you full responsibility over your own funds, in self-custodying, you are often your own biggest threat.

- **No single point of failure:** In a single-sig wallet, if your seed phrase is compromised (without a passphrase), you lose everything. In multi-sig, an attacker needs to find multiple seed phrases in different locations.
- **Shared control:** Ideal for business partners or families. No one person can run off with the funds without the other's consent.
- **Physical protection:** You can keep one key at home, one in a safe deposit box, and one with a trusted lawyer/friend. A house fire or a burglary at one location won't result in lost funds.

Take extreme caution after you have mastered self-custody with a single-signature when considering progressing to multi-signature. Unless you are a well-known individual, or an organisation, or have multiple claims from other parties to your funds in addition to yourself, the author would not recommend a multi-sig setup for self-custodying for the average Joe.

Transacting with Bitcoin

Once sovereignly custodying and securing your bitcoins, you will also be able to sovereignly transact with them.

Let's explore different types of transactions, a general structure, and best practises.

On-Chain Transactions

This section of the manual focuses on the "base layer" of Bitcoin. While the Lightning Network is like a fast-moving stream for payments relatively small in size, on-chain transactions are the heavy-duty freight trains of the network, slightly slower, but incredibly secure and permanent. It is analogous to sending a mail, where you need to specify the exact address, the amount of funds, and pay a tiny fee (the stamp) for it to be delivered to the recipient.

How to Receive Bitcoin On-Chain

Receiving is the easiest way to start. From experience, most of the receiving is done in the context of withdrawing funds from a seller of bitcoin (i.e. an exchange). Below are the typical procedures.

1. Open your wallet and click “Receive”.
2. Generate an Address: The wallet will show a QR code and a string of characters.
 - It’s normal for this address to change every time you receive money; this is a privacy feature! It is like when you receive a piece of gold nugget into your pouch, it occupies another space, and doesn’t amalgamate with the existing nuggets.
3. Share it: Copy-paste the address to the sender, or share the QR code. Best practice, also double check the last 6 letters of the pasted address against the copied address.

How to Send Bitcoin On-Chain

Sending requires more care because mistakes cannot be undone. As soon as a transaction is broadcast to the Bitcoin network, it’s propagated to nodes and miners and cannot be reverted or cancelled. This is a fundamental property of Bitcoin’s decentralised blockchain: no central authority can intervene to stop a transaction. Below are some typical procedures.

1. Open your wallet and click “Send”.
2. Copy the recipient’s address: Never type it out manually. Use a QR scanner or the copy-paste function. Below is a typical Bitcoin address and its corresponding QR code.
Anyone who sees the address or the QR code can engage in direct economic transactions with the owner of the address. Example below.



bc1qdt8y8jv5gmlmfal5f8f7xvqevtcz4xg6a95f3s

3. Choose the Amount: You can usually toggle between the BTC amount and your local currency (e.g., USD).
4. Set the Fee: Your wallet will ask if you want it sent fast, medium, or slow. A higher fee prioritises your transaction, like a mail with a more expensive stamp.
5. Verify & sign: If using a hardware wallet, look at the device screen to make sure the address matches what is on your computer. Press “sign” or “confirm” to sign the transaction with your private key.

6. Broadcast the transaction: Some interfaces will require you to broadcast the transaction to the Bitcoin network. Once the transaction is broadcast to the Bitcoin network, it is irreversible.

Understanding Fees

On-chain fees are not based on the amount of money you send. Sending \$1 million costs close to the same as sending \$1. Instead, you pay for the space your transaction takes up on the Bitcoin ledger.

- Fees are measured in sat/vB (Satoshis per virtual Byte).
- Think of a block in Bitcoin's blockchain like a bus with limited seats. If the bus is full, people will bid higher prices to get a seat on the next trip. Buses arrive and depart periodically on average every 10 minutes.

On-Chain Transacting Tips

- Test small rule: If you are sending a significant amount for the first time, send a tiny "test" amount first to ensure you have the process and the correct recipient address.
- Check the Mempool: You can visit websites like mempool.space to see how busy the network is, or the line for the next bus. If the station is not busy, you can set a very low fee and save money. Only do not enter your wallet address into public websites like mempool.space, as it might link your IP to that wallet address.
- Be aware of which UTXO you spent from which address, most wallets allow users to select UTXO(s) for sending.

UTXOs & Gold Nuggets

More technically speaking, the amount of BTC pertaining to a Bitcoin address is known as an UTXO (unspent transaction output) on the blockchain. The word "unspent" means that amount has not been spent from that address, hence it is available to be spent by the owner of that address.

As a parallel to gold, Bitcoin shares the similarity of being accounted for in digital weights. A single bitcoin can be thought of as weighted in 100,000,000 digital atoms (known as satoshis, abbreviated sats), and the sizes of bitcoins in the wallet's addresses are known as UTXOs (think of it as a nugget). A piece of bitcoin UTXO one holds is akin to a gold nugget; both are unspent by the holder. In parallel, a nugget of gold (an UTXO of bitcoin) can be as small as a single atom (a single sat), or as big as 100 trillion atoms. And the value of the purse (wallet) is given as the total weight (quantity of sats) of all the gold nuggets (UTXOs) combined.

For security, and frankly, logical reasons, you wouldn't take your biggest gold bar (bitcoin UTXO) with you to the market for transacting. It is as if handing the barista a \$1,000,000 note for a \$5 coffee purchase, you might get weird looks as they hand you back \$999,995 in change, and you have exposed yourself to the barista how 'well-off' you are. Therefore, it is highly

advisable to choose which UTXO(s) of bitcoin to spend from when making an on-chain transaction.

Lightning Transactions

This section is arguably the “fun” part of the manual relative to previous sections focusing on storage and security, it will be about Bitcoin being digital cash in your pocket. The Lightning Network is a technological enhancement to Bitcoin that positively transforms it from a slower-moving first-layer money, like shipping physical gold to a second-layer currency moving at near lightspeed with instantaneous settlement.

However, the key difference between BTC and the Lightning network with the layered monetary system of gold and gold-backed paper notes is that the BTC on the first and second layers remains indistinguishable. It is akin the ability to convert any quantities of gold down to the exact unit of atoms into an electronic form, and zapping those e-Atoms across the globe at the speed of light, and the recipient able to convert the e-Atoms back into physical gold anytime at free will, or use it for further Lightning transactions; unlike gold and paper notes, bitcoins remains intrinsically same on both layers.

The Lightning Wallet

Bitcoin’s Lightning Network is typically integrated into soft wallets and is readily available on mobile phones, as it was designed for small and frequent transactions typically paid with bank cards. It is best to think of a Lightning wallet as giving your mobile phone the function of a digital wallet.

To set up for spending bitcoins via the Lightning Network, you will need to fund your Lightning-integrated soft wallet. This is typically done through an on-chain transaction from your non-Lightning wallet (cold or soft) to the Lightning-integrated soft wallet.

There are a few key concepts to know before transacting with the Lightning Network:

- **Sats (Satoshis):** On Lightning, we use Sats as the unit of account instead of BTC, as transactions are typically small in value. There are 100 million sats in one bitcoin. A cup of coffee might be 5,000 sats.
- **Invoices expire:** Unlike a regular Bitcoin address, a Lightning invoice is typically valid for only a few minutes or hours. If it expires, new ones can be generated at any moment.
- **Liquidity (the wiring):** If you have a non-custodial wallet, you might occasionally see a small fee to “open a channel”. This is just the cost of connecting your wallet to the global Lightning Network. Most modern wallets handle this automatically.

Sending And Receiving via Lightning

Sending and receiving in lightning transactions are very similar to on-chain, with the following differences:

- Lightning don’t usually use permanent addresses, but rather invoices

- Lightning transactions doesn't require UTXO management
- Lightning transactions are near instant, with typically lower fees
- Designed for small, frequent transactions and fast payments
- Done mostly through QR codes

How to Receive via Lightning

1. Click "Receive" in your wallet
2. Set the amount for the invoice by typing in how much you want to receive
3. Share the QR code generated on your phone with the other person
4. As soon as the other person has paid, the wallet will notify you, and the sats you receive are available for on-chain and Lightning transactions immediately

How to Send via Lightning

1. The person you are paying will show you a QR code (an invoice)
2. In your Lightning-integrated wallet, hit "Send" and scan their QR code
3. Your wallet will show the amount (e.g., 5,000 sats). You hit "Pay" to confirm

Obtaining Bitcoin

This section concerns the moment you trade your local fiat currency for bitcoins. Because every country has different laws, and every individual's circumstances are unique, this section focuses more on the framework and good practices for acquiring bitcoins, than on a specific approach.

Buying bitcoin is the only process in which you will encounter intermediaries, meaning third parties may be able to restrict what you can and cannot do with your money. Whenever you encounter difficulties in obtaining bitcoins with fiat, know that this is never the fault of Bitcoin, as any difficulty in obtaining of bitcoin with fiat is a testament to how fiat is ultimately a tool for control and restriction, with the ability to allow and restrict the purchase of certain things at the authority of the issuer of the fiat, and whoever custodies your fiat.

The Two Paths: KYC vs Non-KYC

There are two main ways to get Bitcoin. Think of the first as a convenient way, and the second as a private way.

The convenient way, being centralised KYC exchanges, is the most common path, through using a regulated exchange.

- KYC (Know Your Customer): You must provide your national ID and a selfie to create an account. However, hardcore Bitcoiners refer to KYC as "kill your customer", as exchanges are centralised counterparties and honeypots of personal information data, which are vulnerable to coercion from a higher authority, or data breaches of the personal information of their users.

- The trade-off: KYC exchanges are easy to use, but the exchange keeps a permanent record of every satoshi you buy, and may decide to freeze your funds or give away your personal information at their discretion (though not saying KYC exchanges are inherently malicious).
- Crypto-casino: KYC exchanges most often offer access to alternative cryptocurrencies, which are the brainchilds of centralised crypto venture capital projects destined to fail and provide no utility against the mark of the beast.

The private way, is buying directly from other people using online peer-to-peer platforms. Great tutorials and education for this method can be found in BTCsessions' YouTube tutorials and Plan B Academy.

- The trade-off: No personal identification is required, which protects your privacy. However, it has a steeper learning curve and requires more knowledge and “street smarts” to deal directly with a seller.

Good Practices

Withdrawal immediately: The most dangerous thing a dummy can do is leaving their funds on an exchange. Remember, not your keys, not your coins! An exchange is analogous to a public toilet; it is designed to be convenient and very useful when you need it, but once you are done with using it, you should leave as fast as possible. As soon as you buy your bitcoins, withdraw them to your wallet where you control the keys.

Beware of phishing: You may receive a message or email that sounds exactly like “Customer Support” or even a friend asking for your login. Ignore. Legitimate Bitcoin companies will never contact you to ask for your password, additional personal information, or your seed phrase.

Enable 2FA: Use an Authenticator App (like Google Authenticator) to secure your exchange account, if possible, to increase your account security.

Dollar Cost Averaging (DCA): Don't try to guess when the price is “low”. Successful Bitcoiners buy a small, set amount regularly, regardless of price (e.g. \$20 every week). This removes stress and builds your funds steadily over time.

Perform a small test: Buy a tiny amount first and practice withdrawing it to your wallet, to test the rails and familiarise yourself with the process.

Bitcoin Security Cheat Sheat

	NEVER	ALWAYS
Wallets & Self-custody	• Never use custodial wallets for long-term storage • Never pick your own 12 or 24 words seed phrase • For firmware updates of the hardware wallet, never click on links from any wallet/crypto-related email, but instead go to the manufacturer's website, and verify the PGP signatures for updates	• Always generate your seed offline randomly on the hardware wallet • Always only keep a small spending amount in hot wallets • Always buy the hardware wallet directly from the manufacturer, and if possible, order to a PO Box, or work address • Always use brand new factory-made hardware wallets
Seed Phrase Management	• Never store your seed phrase digitally or online, or type it on an external device • Never share your seed phrase with anyone else • Never store all seed backups in one location • Never store the seed phrase together with passphrase	• Always write seed phrases on paper and preferably metal plates (stainless steel) for durability • Always store copies in secure, hidden, separated locations • Always test recovery with insignificant funds (factory reset hard wallet) so you know your backups work • Always try to commit your seed phrase to memory (once doing all the above)
Passphrases	• Never store the passphrase together with seed phrase • Never forget your passphrase; if lost, funds behind the passphrase are irretrievable	• Always store it separately from your seed backups • Always try to commit your passphrase to memory
Transacting with Bitcoin	• Never send large amounts without testing first with a small amount • Never send without double-checking the pasted address with the copied address • Never make transactions from public WiFis or malware-infected devices	• Always transact through soft wallets; hard wallets are purposed exclusively for storage • Always double-check the destination address before sending • Always verify all addresses on the hardware wallet device screen • Always start with sending small test amounts for new recipients or large transactions • Always use trusted nodes and networks, and avoid sending via public WiFi without a VPN

Obtaining Bitcoin	• Never leave any amount of funds on exchanges indefinitely • Never provide any personal information or your seed phrase to “emails” or “requests” from any exchange	• Always buy from reputable exchanges or OTC services with good security records • Always transfer funds to your wallets immediately after purchase • Always enable 2FA with exchanges with an authenticator app • Always assume all forms of contact from exchanges are scams
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Bitcoin Education Resources

A non-exhaustive list of Bitcoin education resource, note that the time you are reading this, all mentioned resources, wallet software and hardware, and/or their providers may have changed or be outdated. Doing your own research before trusting anyone is highly encouraged.

Heck, even Bitcoin itself may or may not have changed from any possible future forks in the software! But ethos of Bitcoin never will: an open, non-discriminatory, permissionless, decentralised, peer-to-peer network for the storage and transfer of value across time and space, a tool to separate money from the state and combat economic disenfranchisement, whether now, or during the time of the Antichrist.

Bitcoin Education

- **Zero2Hero:** An educational website with practical, step-by-step video tutorials and structured courses designed to guide users from their first purchase to advanced self-custody and privacy techniques.
- **Plan B Academy:** A comprehensive educational platform dedicated to teaching Bitcoin, offering certified courses on Bitcoin’s economics, technology, and application, and tutorials on custodying, transacting, and buying. The Plan B Academy expands on all the points covered in this chapter, it is like a theme park for Bitcoin learning!
- **Saylor Academy:** a non-profit organisation established in 1999 that provides free, self-paced, online, and open university-level courses to learners worldwide. It offers specialised, free Bitcoin education, including courses like “Bitcoin for Everybody”.
- **Satoshi Nakamoto Institute:** A library and archive of essential cypherpunk literature, early Bitcoin communications, and technical papers to promote understanding of decentralised, peer-to-peer money.

Further Readings

- **The Bullish Case for Bitcoin** by Vijay Boyapati
- **Layered Money: From Gold and Dollars to Bitcoin and Central Bank Digital Currencies** by Nik Bhatia
- **The Bitcoin Standard** by Saifedean Ammous
- **Check Your Financial Privilege** by Alex Gladstein
- **Broken Money** by Lyn Alden

Final Word from the Author

Congratulations, you made it to the end! Wow... I mean, thank you. Thank you, from the bottom of my heart. I don't know if this book will materially change the world in any way, especially when the Antichrist comes, only the Lord knows. But may it be acceptable to you, and to the Lord. For by the grace of God, I am what I am, and I have written when I have written. May His grace unto me not turn out to be vain, but on the contrary, I have written and laboured more abundantly out of my heart. Yet not I, but the grace of God which is with me. And may the God of peace Himself preserve your whole being completely, whether today, or during the times of the Antichrist, until the day of His return.

And do not be afraid of the Antichrist, for Christ has through death defeated death with the power of His resurrection and life, and if the world has received salvation through His death, much more will we be saved in His life. And He has promised to return to give all life to all who have received Him, to those who are living and those who are asleep. For if Christ does not return, the Antichrist will not come; if the Antichrist does not come, this book would not have been written.

The temporary solution to the mark of the beast may be Bitcoin, but the eternal solution is simply believing into Jesus and receiving Him as your Lord and Saviour, that all of your sins will be forgiven when you repent because Jesus Christ has already bore them on the cross and died for us, through Jesus and Jesus only can we be saved eternally and receive eternal life.

May you receive the true Christ by simply opening more of your heart to Him and receive Him as your Lord and Saviour, and the forgiveness of all of your sins. Now is the time of salvation! May the love of God, the grace of Christ, and the fellowship of the Holy Spirit be with you from now, until the return of our Lord Jesus Christ, until all eternity. Amen!

Sincerely,
The author.